

Parameter determination of the tuned mass damper mitigating the vortex-induced vibration in bridges

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ABSTRACT

This paper investigates the parameter determination of the tuned mass damper (TMD) for mitigating the vortex-induced vibration (VIV) in bridges. At present, there is no consensus on design formulas for the TMD in the VIV mitigation. An equivalent damping ratio is proposed to characterize the TMD contribution to the VIV mitigation based on a semi-empirical nonlinear model. The effects of the uncertainties in aerodynamic parameters and structural parameters on the control effect of the TMD are studied. Three typical design formulas for the TMD are compared from the perspective of the effectiveness, robustness, and reliability. The results show that the equivalent damping ratio depends only on the TMD parameters and is sensitive to the uncertainty in the stiffness of the bridge. The vortex-induced amplitude at lock-in is moderate to the uncertainties in aerodynamic parameters. The design formulas for the free-vibration suppression are the most robust option that avoids a high probability of failure, which are recommended for the TMD mitigating the VIV in bridges. At last, a numerical example of the deep-water non-navigable bridge of the Hong Kong-Zhuhai-Macao Bridge is performed to confirm the rationality of the suggested design formulas.

1. Introduction

Vortex-induced vibration (VIV) generated by vortex shedding behind a bluff body widely exists in the mechanical engineering, civil engineering, marine engineering and aerospace engineering [1]. In bridge engineering, when the vortex shedding frequency approaches to a natural frequency of the bridge, a resonance phenomenon will be locked in a range of wind velocities, which greatly impacts bridge serviceability and driving comfort [2,3]. In addition, the likelihood of the VIV at modest wind velocities is greatly increased with the flexibility of bridges [4,5]. Aerodynamic and mechanical measures are taken to reduce the vortex-induced amplitude of bridges. Since aerodynamic countermeasures may actually result in ambiguous effects on bridges having different sections [6], most bridges adopted the tuned mass damper (TMD) to mitigate the VIV in practice, such as the Trans-Tokyo Bay Crossing Bridge [2], Great Belt East Bridge [7], Rio-Niterói Bridge [8], Osterøy Suspension Bridge [9], Chongqi Bridge [10], and Hong Kong-Zhuhai-Macao Bridge [11,12].

Note that, the determination of TMD parameters greatly depends on the excitation acting on the primary structure and objective function. Over nearly a century, many researchers gave design formulas for the TMD according to different types of excitations, such as the force excitation, base acceleration, free-vibration, self-excitation, earthquake

excitation, and wind load [13–16]. However, just a few studies focused on the determination of TMD parameters for the VIV mitigation of bridges. In [2] and [8], TMD parameters were determined through solving a complex eigenvalue problem in the bridge-TMD system. In the study of Larsen et al. [7], the design formulas for the TMD were analytically obtained based on a three-parameter nonlinear vortex-induced force model. In [10,17], and [18], they all used the Den Hartog's formulas [19] to calculate the optimum TMD parameters. As far, there is neither an in-depth discussion on the difference of these design formulas nor reasonable evaluation on the VIV mitigation of bridges. In order to solve this problem, a reliable mathematical model for the vortex-induced force acting on the bridge should be given. Common mathematical models of the vortex-induced force are the empirical models [20], three-parameter nonlinear model [21], wake oscillator model [22], numerical model [23], Volterra series-based model [3], and so on. Among these models, the semi-empirical nonlinear model [24] is widely used in practice thanks to its simplicity and accuracy. In the lock-in region, the bridge with the semi-empirical nonlinear model can be reduced as a classical van der Pol oscillator [17,25], so that the VIV mitigation of bridges can be considered as the control of a van der Pol oscillator. In this study, the semi-empirical nonlinear model is used to describe the vortex-induced force acting on the bridge. Based on this nonlinear model, an equivalent damping ratio contributed by TMDs is

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Nomenclature

a	vortex-induced amplitude
a_0	vortex-induced amplitude when $\xi_b = 0$
$\hat{a}$	critical vortex-induced amplitude
B	feature dimension of the bridge
C_L	lift coefficient
$\mathbf{F}$	vortex-induced force vector
$F_R(s)$	cumulative distribution function of the resistance
$f_S(s)$	probability density function of the external load
K_L	reduced frequency
L	span length of the bridge
$\mathbf{M}, \mathbf{C}, \mathbf{K}$	mass, damping and stiffness matrices of the bridge
m_b	generalized mass for the fundamental vertical mode of the bridge
m_t	mass of the TMD
m_r	air-bridge mass ratio
P_f	probability of failure
q, q_t	generalized coordinates of the bridge and TM
q_0	steady-state vortex-induced amplitude in the lock-in region
R	resistance
S	external load
S_r	Strouhal number

U	mean wind velocity
x	bridge coordinate across the VIV direction
$\ddot{\mathbf{Y}}, \dot{\mathbf{Y}}, \mathbf{Y}$	acceleration, velocity, and displacement response vectors of the bridge
Y_1, Y_2, ε	aerodynamic parameters
y	vertical displacement of the bridge
Λ_1, Λ_2	complex amplitudes of the bridge and TMD
β	frequency ratio
ξ_b	damping ratio for the fundamental vertical mode of the bridge
ξ_t	damping ratio of the TMD
ξ_e	equivalent damping ratio contributed by TMDs
ξ_{to}	total damping ratio of the bridge
ξ_d	desired damping ratio of the bridge
θ	initial phase
ρ	air density
ϕ	fundamental vertical mode shape
ω	vortex-shedding frequency
ω_b	natural frequency for the fundamental vertical mode of the bridge
ω_t	natural frequency of the TMD
μ	mass ratio
γ	resonance frequency ratio

formulated to illustrate the control effectiveness of TMDs on the VIV mitigation.

The robustness is a major concern in the TMD control considering its serious frequency sensitivity, especially in the small ratio configuration [11]. In the VIV mitigation of bridges, the used mass ratio of the TMD is always small, less than 2%, due to realistic considerations including the installation space and carrying capacity. Thus, the robust design or optimization of TMD parameters is more consequential. Mohtat and Dehghan-Niri [26] proposed a robust design framework of the TMD for the seismic mitigation of building structures. Miguel et al. [27] employed the firefly algorithm to solve the robust optimization of TMD parameters in a vehicle-bridge system. Lievens et al. [28] adopted the multi-interval approach to evaluate the TMD robustness in the vibration serviceability of a footbridge. However, the robust problem of the TMD for mitigating the VIV in bridges is seldom considered seriously. It is challenging to accurately predict the natural frequency and aerodynamic performance of a bridge before it is constructed and in service. Therefore, the designed TMD parameters in the VIV mitigation should be robust enough to the uncertainties in system parameters.

This paper investigates the parameter determination of the TMD for mitigating the VIV in bridges. An equivalent damping ratio is proposed to characterize the TMD contribution to the VIV mitigation based on a semi-empirical nonlinear model. The bridge-TMD system under the vortex-induced force is simplified as a two-degree of freedom (DOF) system, so that the Monte Carlo Simulation (MCS) can be applied to obtain statistical results of the TMD control. The effects of the uncertainties in aerodynamic parameters and structural parameters on the TMD control are studied. Three typical design formulas for the TMD are compared from the perspective of the effectiveness, robustness, and reliability. At last, a numerical example of the deep-water non-navigable bridge of the Hong Kong-Zhuhai-Macao Bridge is performed to confirm the rationality of the suggested design formulas.

2. TMD control of VIV of bridge

2.1. The VIV in bridges

In our study, a semi-empirical nonlinear model [20] is used to simulate the vortex-induced force acting on the bridge, which consists of

the nonlinear aerodynamic damping item, linear aerodynamic stiffness item, and harmonic force item

$$f(\dot{y}, y, t) = \rho U^2 B \left[Y_1(K_L) \left(1 - \varepsilon \frac{y^2}{B^2} \right) \frac{\dot{y}}{U} + Y_2(K_L) \frac{y}{B} + \frac{1}{2} C_L(K_L) \sin(\omega t + \theta) \right] \quad (1)$$

where y is the displacement of the vertical DOF; ρ is the air density; U is the mean wind velocity; B is the feature dimension of the bridge; $K_L = \frac{\omega B}{U}$ is the reduced frequency; Y_1, Y_2 , and ε are the aerodynamic parameters which characterize the negative damping ratio, aerodynamic stiffness, and self-limited amplitude, respectively; C_L is the lift coefficient; ω and θ are the vortex-shedding frequency and the initial phase.

In the lock-in region, the contributions of the aerodynamic stiffness and harmonic force on the vortex-induced amplitude could be neglected [20,24]. Thus, the vortex-induced force can be reduced as

$$f(\dot{y}, y, t) = \rho U^2 B \bar{Y}_1 \left(1 - \varepsilon \frac{y^2}{B^2} \right) \frac{\dot{y}}{U} \quad (2)$$

Since in the lock-in region the vortex-shedding frequency is approximately equal to a natural frequency of the bridge [20,24], $\bar{Y}_1 = Y_1 \left(\frac{\omega_b B}{U} \right)$, where ω_b is the natural frequency for the fundamental vertical mode of the bridge.

The bridge can be modeled as a multiple-DOF system so that the equation of motion of the bridge in the lock-in region can be expressed as

$$\mathbf{M}\ddot{\mathbf{Y}}(t) + \mathbf{C}\dot{\mathbf{Y}}(t) + \mathbf{K}\mathbf{Y}(t) = \mathbf{F}(\dot{\mathbf{Y}}, \mathbf{Y}, t) \quad (3)$$

where $\ddot{\mathbf{Y}}, \dot{\mathbf{Y}}$, and $\mathbf{Y}$ are the acceleration, velocity, and displacement response vectors of the bridge, respectively; $\mathbf{M}, \mathbf{C}$ and $\mathbf{K}$ are the mass, damping and stiffness matrices of the bridge; $\mathbf{F}$ is the vortex-induced force vector.

Since the vortex-induced resonance in bridges is always dominated by the fundamental vertical mode, i.e., $\mathbf{Y} = \phi q$, the generalized equation of motion can be expressed as

$$\ddot{q} + 2\omega_b \xi_b \dot{q} + \omega_b^2 q = \frac{1}{m_b} \rho U B \bar{Y}_1 (\alpha_2 - \bar{\varepsilon} q^2) \dot{q} \quad (4)$$

where ϕ is the fundamental vertical mode shape; q is the generalized

coordinate; $m_b = \phi^T \mathbf{M} \phi$ and ξ_b are the generalized mass and damping ratio for the fundamental vertical mode of the bridge; $\alpha_2 = \int_0^L \phi^2(x) dx$, $\alpha_4 = \int_0^L \phi^4(x) dx$, and $\bar{\varepsilon} = \frac{\varepsilon}{\beta^2}$; L is the span length of the bridge; x is the bridge coordinate across the VIV direction.

Furthermore, the system presented in Eq. (4) can be written in a van der Pol oscillator [17,24,25]

$$\ddot{q} + \left(2\omega_b \xi_b - \frac{\rho U B \bar{Y}_1}{m_e} + \frac{\bar{\varepsilon} \alpha_4 \rho U B \bar{Y}_1}{\alpha_2 m_e} q^2 \right) \dot{q} + \omega_b^2 q = 0 \quad (5)$$

where $m_e = \frac{m_b}{\alpha_2}$ is defined as the effective mass in the VIV. To analyze the effects of various parameters on the VIV response, the method of slowly varying parameters [20] is adopted to obtain approximate steady-state amplitude,

$$q_0 = 2 \sqrt{\frac{\alpha_2}{\bar{\varepsilon} \alpha_4}} \cdot \sqrt{1 - \frac{2m_e \omega_b \xi_b}{\rho U B \bar{Y}_1}} \quad (6)$$

where q_0 is the steady-state vortex-induced amplitude in the lock-in region. From Eq. (6), it is easy to observe that the VIV of the bridge can be completely mitigated when $\xi_b \geq \frac{\rho U B \bar{Y}_1}{2m_e \omega_b}$.

2.2. TMD control

In order to improve the damping ratio of the bridge, the TMD is installed in the bridge, as shown in Fig. 1. The equations of motion of the bridge-TMD system can be expressed as

$$\ddot{q} + 2\xi_b \omega_b \dot{q} + \omega_b^2 q = \frac{1}{m_b} \rho U B \bar{Y}_1 (\alpha_2 - \bar{\varepsilon} \alpha_4 q^2) \dot{q} - \mu (\ddot{q} + \ddot{q}_t) \quad (7)$$

$$\ddot{q}_t + \ddot{q} + 2\omega_t \xi_t \dot{q}_t + \omega_t^2 q_t = 0 \quad (8)$$

where q_t is the generalized coordinate of the TMD; ω_t and ξ_t are the natural frequency and damping ratio of the TMD; $\mu = \frac{m_t}{m_b}$ is the mass ratio; m_t is the mass of the TMD.

According to the Krylov Bogoliubov method [17], the bridge-TMD system under the VIV can be assumed to be in the quasi-steady state

$$q(t) = \Lambda_1 e^{i\omega t} \quad (9)$$

$$q_t(t) = \Lambda_2 e^{i\omega t} \quad (10)$$

where Λ_1 and Λ_2 are the complex amplitudes of the bridge and TMD.

Substituting equations (9) and (10) into equation (8) yields

$$\frac{\Lambda_2}{\Lambda_1} = g + hi = \frac{\gamma^2(\beta^2 - \gamma^2)}{(\beta^2 - \gamma^2)^2 + (2\xi_t \beta \gamma)^2} - \frac{\gamma^2(2\xi_t \beta \gamma)}{(\beta^2 - \gamma^2)^2 + (2\xi_t \beta \gamma)^2} i \quad (11)$$

where $\beta = \frac{\omega}{\omega_b}$ is the frequency ratio; $\gamma = \frac{\omega}{\omega_b}$ is the resonance frequency ratio. According to $\ddot{q}_t(t) = -\omega^2 g q - \omega h \dot{q}$, the inertial force provided by the TMD can be considered as the sum of the displacement feedback and velocity feedback. Furthermore, Eq. (7) can be rewritten as follows

$$f(q, \dot{q}) = \frac{\rho U B \bar{Y}_1}{m_b} (\alpha_2 \dot{q} - \bar{\varepsilon} \alpha_4 q^2 \dot{q}) + \mu (\omega^2 g q + \omega h \dot{q}) + \mu \omega^2 q - 2\xi_b \omega_b \dot{q} \quad (12)$$

Assuming that $q = a \cos(\omega t - \psi)$ and $\dot{q} = -\omega a \sin(\omega t - \psi)$, a and ψ are the vortex-induced amplitude and phase slowly varying with time, which are defined by

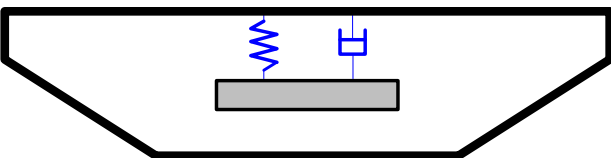


Fig. 1. The bridge-TMD system.

$$\dot{a} = \frac{1}{2\pi\omega} \int_0^{2\pi} [f(q, \dot{q}) + a(\omega^2 - \omega_b^2) \cos\tau] \sin\tau d\tau \quad (13)$$

$$\dot{\psi} = \frac{1}{2\pi\omega a} \int_0^{2\pi} [f(q, \dot{q}) + a(\omega^2 - \omega_b^2) \cos\tau] \cos\tau d\tau \quad (14)$$

The condition of the steady-state solution is written as follows

$$\dot{a} = \frac{da}{dt} = 0 \quad (15)$$

$$\dot{\psi} = \frac{d\psi}{dt} = 0 \quad (16)$$

According to Eqs. (15) and (16), the two equations can be obtained separately

$$\frac{\rho U B \bar{Y}_1}{4m_b} \bar{\varepsilon} \alpha_4 a^2 + 2\xi_b \omega_b - \mu \omega h - \frac{\rho U B \bar{Y}_1}{m_b} \alpha_2 = 0 \quad (17)$$

$$\gamma^6 - (2\beta^2 - 4\xi_t^2 \beta^2 + 1 + \mu\beta^2 - 4\mu\xi_t^2 \beta^2) \gamma^4 + (\beta^4 + 2\beta^2 - 4\xi_t^2 \beta^2 + \mu\beta^4) \gamma^2 - \beta^4 = 0 \quad (18)$$

According to Eq. (17), the vortex-induced amplitude of the bridge is given by

$$a = 2 \sqrt{\frac{\alpha_2}{\bar{\varepsilon} \alpha_4}} \cdot \sqrt{1 - \frac{2m_e \omega_b \left(\xi_b + \frac{\mu \gamma^4 \xi_t \beta}{(\beta^2 - \gamma^2)^2 + (2\xi_t \beta \gamma)^2} \right)}{\rho U B \bar{Y}_1}} \quad (19)$$

where γ is calculated by Eq. (18). Note that, the role of the TMD is equivalent to increasing the damping ratio of the bridge in the VIV mitigation, through the added equivalent damping ratio, $\xi_e = \frac{\mu \gamma^4 \xi_t \beta}{(\beta^2 - \gamma^2)^2 + (2\xi_t \beta \gamma)^2}$.

3. Uncertainty analysis

In the VIV evaluation, the vortex-induced amplitude of the bridge in the lock-in region a is often selected as the performance index. The VIV mitigation is achieved when satisfying $\hat{a} \geq a$, where $\hat{a}$ is the critical vortex-induced amplitude, which is determined according to the guidance, code, and design requirement. To give a general result, the dimensionless total damping ratio of the bridge, $\xi_{to} = \xi_b + \xi_e$, is used as the performance index instead of a in our study. Substituting Eq. (19) into $\hat{a} \geq a$ yields,

$$\xi_{to} \geq \frac{\rho U B \bar{Y}_1}{2m_e \omega_b} \left(1 - \frac{\bar{\varepsilon} \alpha_4 \hat{a}^2}{4\alpha_2} \right) \quad (20)$$

Introducing $m_r = \frac{\rho B^2}{m_e}$, $S_r = \frac{\omega_b B}{2\pi U}$, and $a_0 = 2 \sqrt{\frac{\alpha_2}{\bar{\varepsilon} \alpha_4}}$, Eq. (20) can be rewritten as

$$\xi_{to} \geq \frac{m_r \bar{Y}_1}{4\pi S_r} \left(1 - \frac{\hat{a}^2}{a_0^2} \right) \quad (21)$$

where m_r is the air-bridge mass ratio; S_r is the Strouhal number in the lock-in region; a_0 is the vortex-induced amplitude in the lock-in region when the damping ratio of the bridge is zero, as presented in Eq. (6) or Eq. (19). According to Eq. (21), the left item and right item can be considered as the control effect and control objective, respectively. $\xi_d = \frac{m_r \bar{Y}_1}{4\pi S_r} \left(1 - \frac{\hat{a}^2}{a_0^2} \right)$ is defined as the desired damping ratio of the bridge, which is dependent on the bridge parameters, aerodynamic parameters, and critical vortex-induced amplitude.

3.1. Uncertainty of control objective

Experimental and numerical results showed that the aerodynamic parameters ($\bar{Y}_1$ and ε) identification are sensitive to the vortex-induced amplitude data [24,25]. The Strouhal number S_r reflects the occurring condition of the VIV, which is greatly affected by the Reynolds number [29]. Therefore, $\bar{Y}_1$, S_r , and ε (or $\frac{1}{a_0^2}$) can be considered as random

variables. Although there are no statistics for their uncertainties, we assume that three random variables all obey the normal distribution, which is often used in the natural science and social science to represent real-valued random variables whose distributions are not known [30,31].

$$Y_1 \sim \mathcal{N}(\mu_{Y_1}, \sigma_{Y_1}^2) \quad (22)$$

$$S_r \sim \mathcal{N}(\mu_{S_r}, \sigma_{S_r}^2) \quad (23)$$

$$\frac{1}{a_0^2} \sim \mathcal{N}\left(\frac{\mu_{1/a_0^2}}{a_0^2}, \frac{\sigma_{1/a_0^2}^2}{a_0^2}\right) \quad (24)$$

where μ_{Y_1} , μ_{S_r} , and $\frac{\mu_{1/a_0^2}}{a_0^2}$ are the means of three variables; σ_{Y_1} , σ_{S_r} , and

$\frac{\sigma_{1/a_0^2}}{a_0^2}$ are the standard deviations of three variables. $z = \frac{Y_1}{S_r}$ is introduced to reduce the number of variables in the robust analysis. The distribution of the uncorrelated normal ratio can be approximately by a log transformation,

$$z = \frac{\mu_{Y_1} + \mathcal{N}(0, \sigma_{Y_1}^2)}{\mu_{S_r} + \mathcal{N}(0, \sigma_{S_r}^2)} = \frac{\mu_{Y_1} + Y_1}{\mu_{S_r} + S_r} = \frac{\mu_{Y_1}}{\mu_{S_r}} \frac{1 + \frac{Y_1}{\mu_{Y_1}}}{1 + \frac{S_r}{\mu_{S_r}}} \quad (25)$$

After a log transformation, we get

$$\ln(z) = \ln\left(\frac{\mu_{Y_1}}{\mu_{S_r}}\right) + \ln\left(1 + \frac{Y_1}{\mu_{Y_1}}\right) - \ln\left(1 + \frac{S_r}{\mu_{S_r}}\right) \quad (26)$$

According to the Taylor series expansion, $\ln(1 + \theta) = \theta - \frac{\theta^2}{2} + \frac{\theta^3}{3} - \dots + (-1)^{n+1} \frac{\theta^n}{n}$. $\ln(z)$ can be approximated by

$$\ln(z) \approx \ln\left(\frac{\mu_{Y_1}}{\mu_{S_r}}\right) + \frac{Y_1}{\mu_{Y_1}} - \frac{S_r}{\mu_{S_r}} - \mathcal{N}\left(\ln\left(\frac{\mu_{Y_1}}{\mu_{S_r}}\right), \frac{\sigma_{Y_1}^2}{\mu_{Y_1}^2} + \frac{\sigma_{S_r}^2}{\mu_{S_r}^2}\right) \quad (27)$$

Then, we can obtain the arithmetic mean and variance of a log-normally distributed variable z ,

$$\mu_z = e^{\ln\left(\frac{\mu_{Y_1}}{\mu_{S_r}}\right) + \frac{1}{2}\left(\frac{\sigma_{Y_1}^2}{\mu_{Y_1}^2} + \frac{\sigma_{S_r}^2}{\mu_{S_r}^2}\right)} = \frac{\mu_{Y_1}}{\mu_{S_r}} e^{\frac{1}{2}\left(\frac{\sigma_{Y_1}^2}{\mu_{Y_1}^2} + \frac{\sigma_{S_r}^2}{\mu_{S_r}^2}\right)} \quad (28)$$

$$\sigma_z^2 = \left(\frac{\mu_{Y_1}}{\mu_{S_r}}\right)^2 e^{\left(\frac{\sigma_{Y_1}^2}{\mu_{Y_1}^2} + \frac{\sigma_{S_r}^2}{\mu_{S_r}^2}\right)} \left(e^{\left(\frac{\sigma_{Y_1}^2}{\mu_{Y_1}^2} + \frac{\sigma_{S_r}^2}{\mu_{S_r}^2}\right)} - 1\right) \quad (29)$$

Finally, the mean and coefficient of variation (CV) of the control objective characterized by ξ_d can be calculated according to the rules for expectations of independent random variables.

3.2. Uncertainty of control effect

It is easy to observe from Eqs. (18) and (19) that the system parameters affect together the control effect of the TMD. Compared to the mass, the stiffness and damping ratio are more easily affected by various disturbances, to have non-negligible uncertainty. Besides, since the damping ratio of flexible bridges is very small, less than 1% [11], the effects of itself and its uncertainty can be ignored in the robust analysis. The GERB CHINA Company tested their spiral steel springs used in TMDs under 3.05×10^6 sinusoidal load cycles [11]. The result showed that the change of the spring stiffness was less than 0.5%, indicating that the uncertainty in the TMD's stiffness also can be ignored. Therefore, only k_b and ξ_t are considered as random variables obeying the normal distribution.

It is difficult to analytically obtain the mean and CV of the equivalent damping ratio, because the expression of Eq. (19) is complex, and a cubic equation also needs to be solved. Considering that

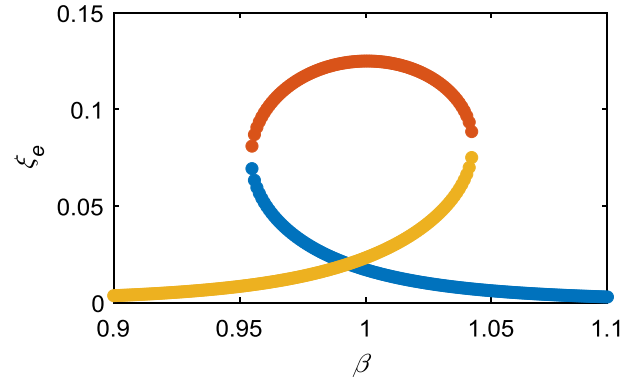


Fig. 2. Equivalent damping ratio when $\mu = 0.01$ and $\xi_t = 0.02$.

there are only two random variables, the MCS with high accuracy is used to calculate the mean, variance, and probability distribution of the equivalent damping ratio of the bridge, ξ_e . Note that, Eq. (18) is a classical cubic equation, $X^3 + AX^2 + BX + C = 0$, there may be triple positive solutions when ξ_t is very small and $\beta \approx 1$. As a result, an unpredictable jump in the control effect of the TMD will occur in such a combination of parameters case, as shown in Fig. 2. The phenomenon demonstrates that the high equivalent damping ratio is ambiguous and unstable [17]. Based on the finding, the simulation with smallest equivalent damping ratio is finally selected as the true simulation in the robust analysis. Fig. 3 shows the flowchart of the MCS of the control effect.

3.3. Uncertainty result

To illustrate the uncertainties of ξ_d , wind tunnel test results of a rectangular cylinder are used as nominal values of aerodynamic parameters. In wind tunnel tests conducted by Marra et al. [25], $B = 300\text{mm}$, $S_r = 0.136$, $m_r = 0.0011$, $\bar{Y}_1 = 6.27$, and $\varepsilon = 1082.2$. In the identification process of aerodynamic parameters, they found that $\bar{Y}_1$ is very sensitive to the identification method and disturbances, so that the CV of $\bar{Y}_1$ is assumed as 10%. The CVs of S_r and ε are both assumed as 5%. Since the non-dimensional amplitude of the rectangular cylinder is $a_0 = 0.06B$ [25], thus the assumption of $\hat{a} = 0.02B$ is reasonable to show the VIV mitigation in our study.

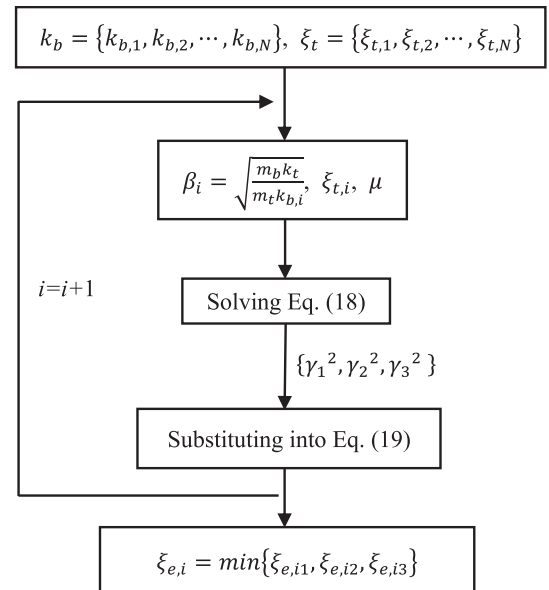
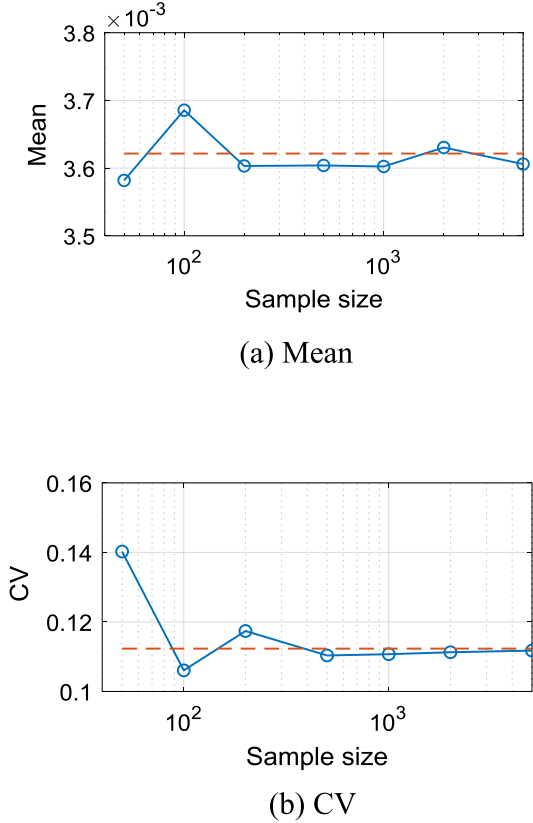
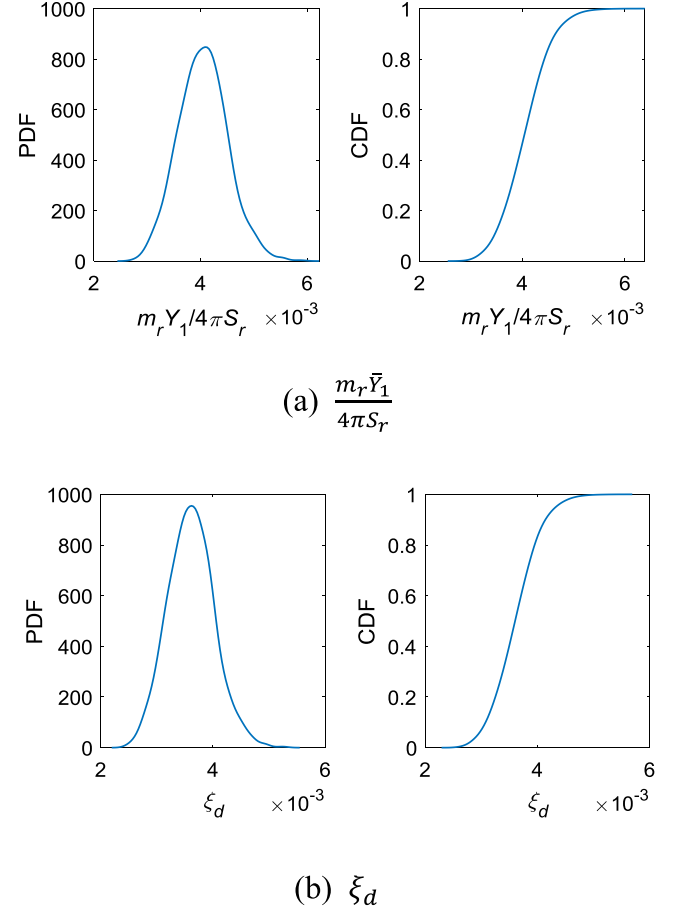


Fig. 3. Flowchart of the MCS of the control effect.

Fig. 4. MCS result of ξ_d increasing sample size.

As a comparison, the MSC is used to estimate the mean, CV, probability density function (PDF), and cumulative distribution function (CDF) of ξ_d . The sample size or the simulation number determines the MSC accuracy, the greater the simulation number is, the greater the statistical reliability is [30]. Fig. 4 shows the MCS result of ξ_d increasing sample size. It can be seen that the mean and CV of ξ_d both keep stable when the sample size is greater than 500. In order to balance the MCS accuracy and computing cost, the sample sizes of S_r , Y_1 , and ε are taken as 2000. Note that, the dotted line refers to the approximate analytical result according to Eqs. (22)–(29). A good match in Fig. 4 indicates that Eqs. (22)–(29) can estimate the mean and CV of ξ_d as well as the MCS. Fig. 5 shows the estimated PDF and CDF of $\frac{m_r \bar{Y}_1}{4\pi S_r}$ and ξ_d . If the VIV needs to be completely mitigated, the desired damping ratio should be equal to $\frac{m_r \bar{Y}_1}{4\pi S_r}$, not ξ_d with a critical vortex-induced amplitude, $\hat{a}$. Since $\frac{\hat{a}^2}{a_0^2}$ is much smaller than 1 in the expression of $\left(1 - \frac{\hat{a}^2}{a_0^2}\right)$, the CV of $\left(1 - \frac{\hat{a}^2}{a_0^2}\right)$ is small compare to that of $\frac{m_r \bar{Y}_1}{4\pi S_r}$. According to rules for expectations of independent random variables, the uncertainty in ξ_d is dominated by the uncertainty in $\frac{m_r \bar{Y}_1}{4\pi S_r}$, so that the CV of ξ_d is approximately equal to that of $\frac{m_r \bar{Y}_1}{4\pi S_r}$. It can be verified that the two PDF curves of $\frac{m_r \bar{Y}_1}{4\pi S_r}$ and ξ_d have the similar curve shape. Due to the introduction of $\hat{a}$, PDF and CDF curves of ξ_d are both shifted to the small-value region compared to those of $\frac{m_r \bar{Y}_1}{4\pi S_r}$.

Assume that a TMD with $\mu = 0.01$, $\beta = 1$, and $\xi_t = 0.05$ is used to mitigate the VIV, the CVs of k_b (or $\frac{1}{\beta^2}$) and ξ_t are assumed as 5% and 10%, respectively. Fig. 6 shows the MCS result of ξ_e increasing sample size. It can be seen that the simulation result can keep stable when the sample size is greater than 1000, so that the sample size is also taken as 2000. Fig. 7 shows the estimated PDF and CDF of ξ_e . It can be seen from Figs. 6 and 7 that the uncertainty in ξ_e is very large, with a CV of 30% that is three times the maximum CV of k_b and ξ_t , which indicates that the control effect is sensitive to these uncertainties. To find the

Fig. 5. Estimated PDF and CDF of $\frac{m_r \bar{Y}_1}{4\pi S_r}$ and ξ_d .

dominant random variable, Fig. 8 shows the MCS result of ξ_e with varying CVs of k_b and ξ_t . It is easy to find that the effect of the uncertainty in ξ_t on the mean and CV of ξ_e is so small that can be neglected. The uncertainty in k_b plays a dominant role in both the mean and CV of ξ_e , the larger the CV of k_b is, the smaller the mean of ξ_e is and the larger the CV of ξ_e is, which also indicates that the robustness of the TMD is bad.

4. Robust analysis

Based on above results, we should select robust TMD parameters to the uncertainties in the desired damping ratio and stiffness of the bridge.

4.1. Design formulas

In the VIV mitigation of bridges or the control of a van der Pol oscillator using the TMD, three types of design formulas are used to calculate TMD parameters, free-vibration type (F-type) [14], harmonic-force type (H-type) [19], and self-excited type (S-type) [14,32], as presented in Table 1.

These design formulas are established based on the linearized system presented in Eqs. (7) and (8). According to the Lyapunov's Linearization Theorem [33], the characteristic equation of the linearized system is given by

$$\lambda^4 + (2\mu\beta\xi_t + 2\beta\xi_t + 2\xi_b)\lambda^3 + (1 + \beta^2 + \mu\beta^2 + 4\beta\xi_b\xi_t)\lambda^2 + (2\beta\xi_t + 2\beta^2\xi_b)\lambda + \beta^2 = 0 \quad (30)$$

where λ is the complex eigenvalue. The F-type TMD aims to maximize the minimum modal damping ratio, the H-type or Den Hartog's

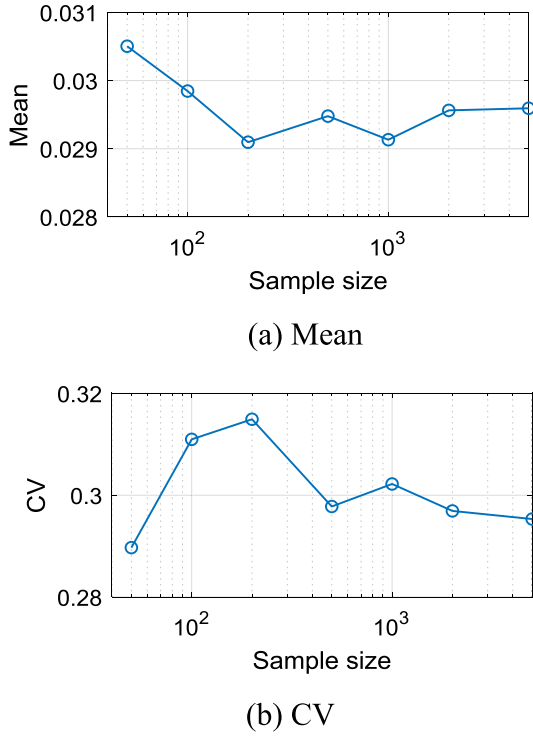


Fig. 6. MCS result of ξ_e increasing sample size.

TMD aims to minimize the peak of the structural displacement response curve, and the S-type aims to make the real parts of two pairs λ be both zero.

Fig. 9 shows the optimal β and ξ_i of the TMD according to the three design formulas. Since the mass ratio is smaller than 2% in VIV mitigation of bridges, the range of μ considered here is from 0.1% to 2%. We find that the maximum difference of the optimal β is less than 1% that can be neglected in practice. However, there is considerable difference in the optimal ξ_i within all the mass ratios, the F-type and S-type formulas give the largest and smallest damping ratio of the TMD, respectively. Fig. 10 shows the mean and CV of ξ_e according to the three design formulas. The dash lines refer to the equivalent damping ratios without any uncertainties. The S-type formulas give the largest equivalent damping ratio within all the mass ratios, with the best control effect, followed by the H-type formulas and F-type formulas. The similar result is observed in the mean of ξ_e , indicating that the S-type TMD has the best effectiveness. However, the results of the mean

and CV of ξ_e indicate that the F-type formulas selection is the most robust within all the mass ratios, followed by the H-type formulas and S-type formulas.

4.2. Reliability-based evaluation

The structural reliability is widely used to measure the structural safety against a failure occurring when the external load S is larger than the resistance R . The probability of failure can be formulated by,

$$P_f = P[R \leq S] = \int_0^\infty f_S(s) \left[\int_0^s f_R(r) dr \right] ds = \int_0^\infty F_R(s) f_S(s) ds \quad (31)$$

where P_f is the probability of failure; $F_R(s)$ is the CDF of the resistance R , and $f_S(s)$ is the PDF of the external load S . Note that, Eq. (31) is established based on the assumption that R and S are independent. Instead, a general formula for the probability of failure is defined as,

$$P_f = \int_{G(\mathbf{x}) \geq 0}^0 f_{\mathbf{x}}(\mathbf{x}) d\mathbf{x} \quad (32)$$

where $\mathbf{x}$ is the vector of random variables in the limit state function $G = S - R$. When the MCS is used to estimate the probability of failure, P_f is approximately given

$$P_f \approx \frac{\sum_{j=1}^N I(\mathbf{x}_j)}{N} \quad (33)$$

where N is the sample size; $I(\mathbf{x}_j)$ is defined as

$$I(\mathbf{x}_j) = \begin{cases} 1 & G(\mathbf{x}_j) \geq 0 \\ 0 & G(\mathbf{x}_j) < 0 \end{cases} \quad (34)$$

In the VIV mitigation of bridges, the random variables are $\mathbf{x} = \{Y_1, S_r, \varepsilon, k_b, \xi_i\}^T$, the external load S and resistance R mean the desired damping ratio ξ_d and equivalent damping ratio ξ_e , respectively. The probability of failure P_f can well consider the balance between the effectiveness and robustness.

Fig. 11 shows the probability of failure of the TMD in the VIV mitigation. The small and large uncertainties mean that the CV vectors of $\mathbf{x}$ are (10%, 5%, 5%, 5%, 10%) and (20%, 10%, 10%, 10%, 20%), respectively. As shown in Fig. 11, the probability of failure of the three TMDs all greatly decreases with the mass ratio, and then will be stable when the mass ratio reaches a critical value, which indicates that the proper mass increase is effective in improving the TMD control. In the small uncertainty case, the F-type TMD has the smallest probability of failure within all the mass ratios, followed by the H-type TMD and S-type TMD. According to the results in Fig. 9(b) and Fig. 11, it can be concluded that increasing the damping ratio of the TMD is equivalent to increasing the mass of the TMD in reducing the probability of failure. Since ξ_i

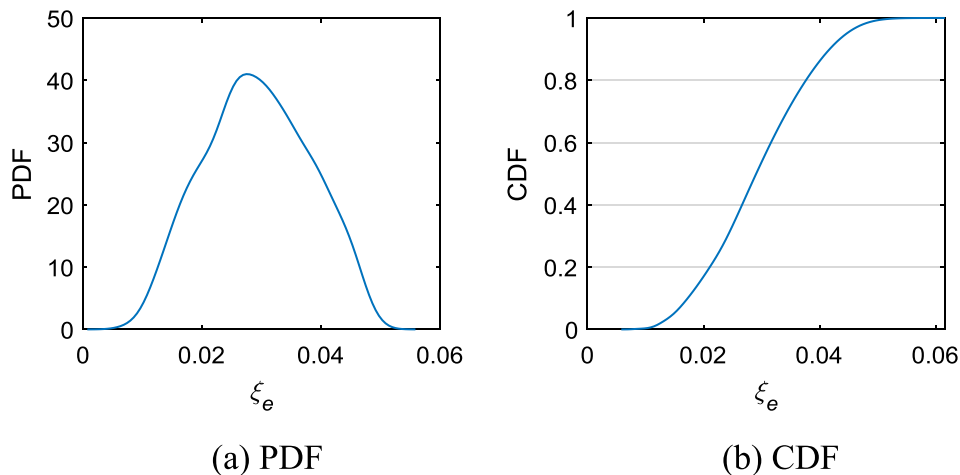


Fig. 7. Estimated PDF and CDF of ξ_e .

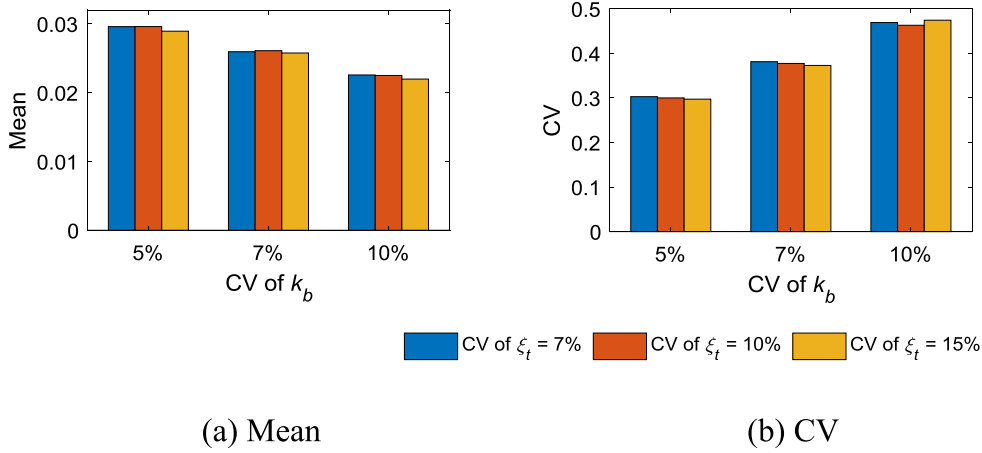
Fig. 8. MCS result of ξ_e with varying CVs of ξ_b and ξ_t .

Table 1
Design formulas of TMD parameters.

Types	Free vibration	Harmonic force	Self-excited
β	$\frac{1}{1+\mu}$	$\frac{1}{1+\mu}$	$\frac{1}{\sqrt{1+\mu}}$
ξ_t	$\sqrt{\frac{\mu}{1+\mu}}$	$\sqrt{\frac{3\mu}{8(1+\mu)}}$	$\sqrt{\frac{\sqrt{1+\mu}-1}{2\sqrt{1+\mu}}}$

calculated by the F-type formula is much larger than those calculated by S-type and H-type formulas, the F-type TMD still has the smallest probability of failure, and the H-type TMD has a slightly smaller probability of failure than the S-type TMD in the large uncertainty case, as shown in Fig. 11(c). Note that, the control objective will be improved from ξ_d to $\frac{m_r \bar{Y}_1}{4\pi S_T}$ when the VIV needs to be completely mitigated. Since the complete VIV mitigation requires a larger equivalent damping ratio, the probability of failure in Fig. 11(b) and (d) is larger than that in Fig. 11(a) and (c), but with the same curve shape. The above results suggest that, the F-type formulas are the best option for TMD parameters considering the effectiveness and robustness simultaneously.

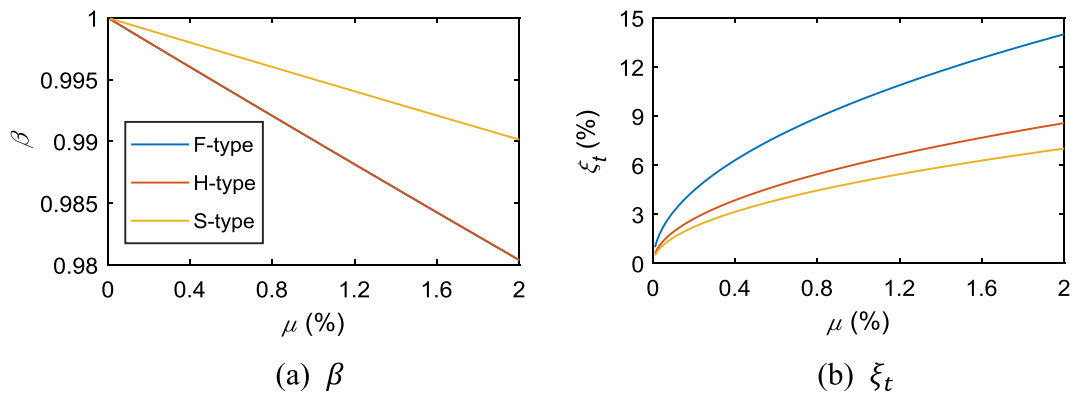
5. Numerical example

The numerical example of a long-span bridge under the VIV is used to illustrate the role of TMD parameters in the VIV mitigation. The model is the deep-water non-navigable bridge of the Hong Kong-Zhuhai-Macao Bridge, a 6×110 m continuous steel box-girder bridge, as shown in Fig. 12. The finite element analysis shows that the

generalized mass and natural frequency for the fundamental vertical mode are 8888.9 tons and 0.806 Hz. The damping ratio of the bridge is assumed as 0.3%.

Wind tunnel test results of the scaled sectional and full-bridge aeroelastic models showed that the 6×110 m bridge will experience a vertical vortex-induced resonance with a maximum amplitude of 140 mm – 150 mm at the wind velocity of 27 m/s – 35 m/s [34]. A simplified decay-to-resonance method [35] is used to identify the aerodynamic parameters, $\bar{Y}_1=15.674$ and $\varepsilon=2.9 \times 10^3$. Fig. 13 shows the vortex-induced amplitude of the bridge in the lock-in region. In order to reproduce the limit cycle phenomenon [3] in the vortex-induced resonance, two representative cases with initial values of 50 mm (smaller than 140 mm) and 300 mm (larger than 140 mm) are considered here. It can be seen that the vortex-induced amplitude of the 6×110 m bridge gradually reaches a stable value 140 mm over time, which indicates that the semi-empirical nonlinear model can capture the limit cycle phenomenon of the VIV, and the initial condition does not affect the steady-state amplitude.

The vortex-induced amplitude of 140 mm exceeds the limit value suggested by China code (JTG/T 3360-01-2018 [36]), $40/f_v = 50$ mm, where $f_v = 0.806$ Hz is the vertical vibration frequency. Wind tunnel test results showed that the vortex-induced amplitude will be reduced to 15 mm when the damping ratio of the bridge is 1% [34]. According to Eq. (20), the desired damping ratio of the bridge is 0.92% to stratify the limit value of 50 mm. The TMD scheme is adopted to mitigate the unwanted VIV. The eight TMDs with a total mass of 32 tons ($\mu = 0.36\%$) were installed in the center of the second and fifth spans. According to Eq. (19) and Table 1, the equivalent damping ratios of the F-type, H-type, and S-type TMDs are 1.5%, 2.4%, and 3.0%, the corresponding total damping ratio are 1.8%, 2.7%, and 3.3%. Fig. 14 shows the control

Fig. 9. Optimal β and ξ_t according to the three design formulas.

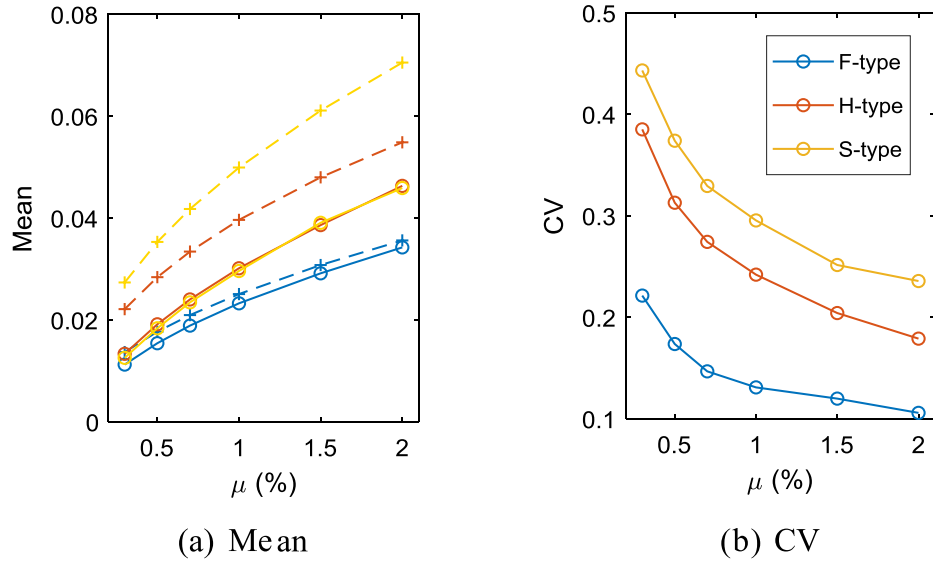
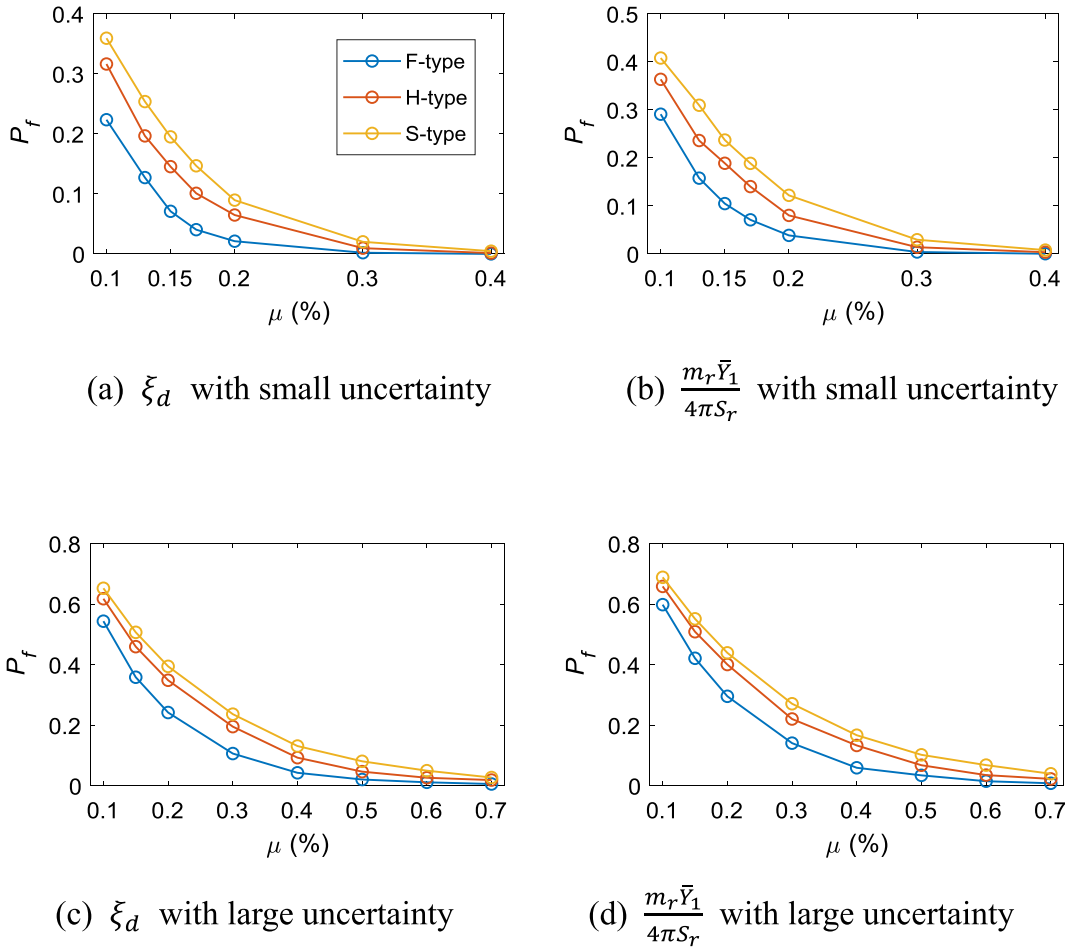
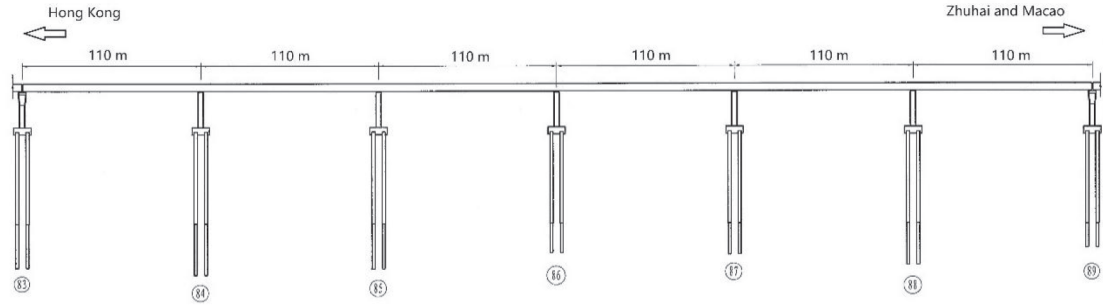
Fig. 10. Mean and CV of ξ_e according to the three design formulas.

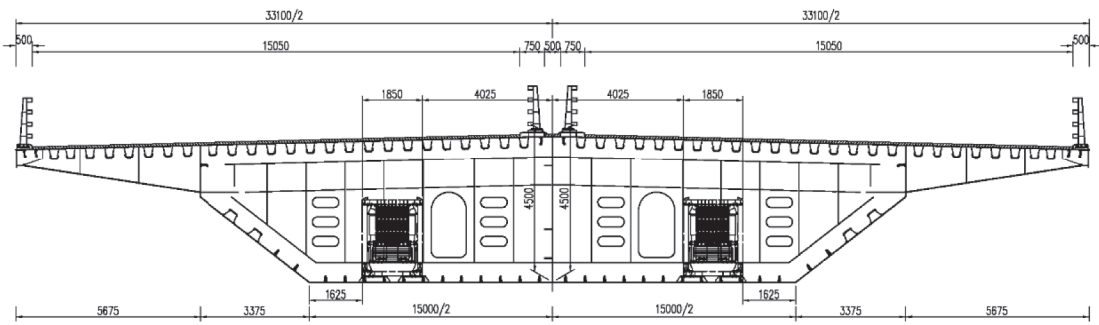
Fig. 11. The probability of failure of the TMD control.

effect of the three types of TMDs. Since the steady-state amplitude of the bridge without TMDs is 140 mm, the initial amplitude of the bridge is assumed as 140 mm to intuitively reflect the control effect of the TMD. It can be seen that the F-type, H-type, and S-type TMDs can mitigate the VIV to satisfy the limit value of 50 mm at 18.63 s, 12.42 s, and 11.18 s, which indicates the S-type TMD has the most optimal control

effect, followed by the H-type TMD and F-type TMD. It is noted that there are no uncertainty in the control effect shown in Fig. 14, in other words, only the effectiveness of the TMD is considered here. Besides, a beat phenomenon decay instead of an exponential decay is observed in the H-type TMD and S-type TMD controls, because the damping ratios of the two TMDs are so small that the bridge-TMD system becomes



(a) A 6×110 m continuous steel box-girder bridge



(b) Girder section

Fig. 12. The deep-water non-navigable bridge of the Hong Kong-Zhuhai-Macao Bridge.

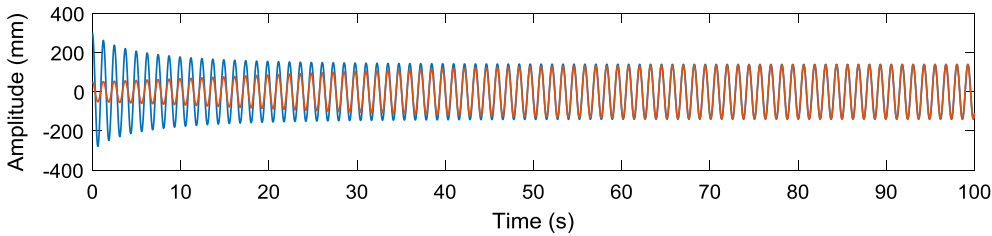


Fig. 13. The vortex-induced amplitude of the bridge in the lock-in region.

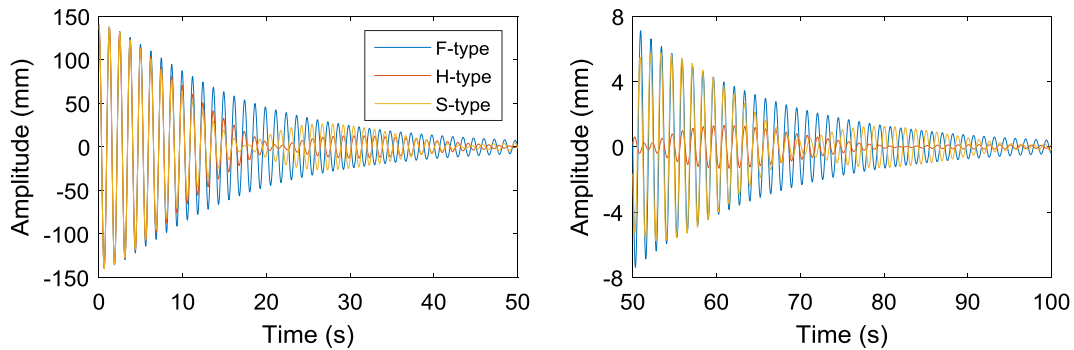


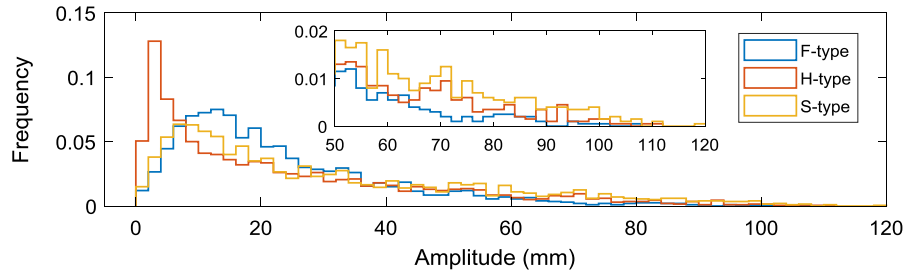
Fig. 14. Control effect of the three types of TMDs.

severely coupled.

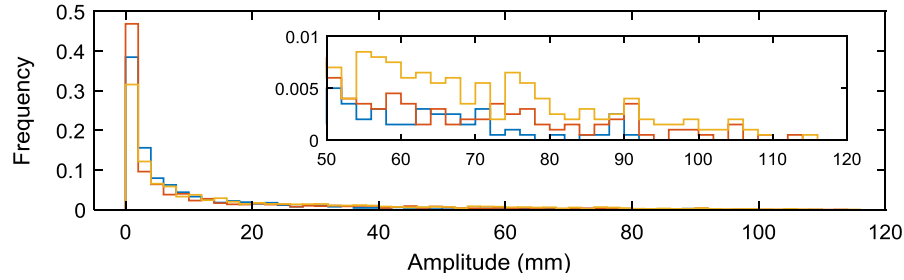
Fig. 15 shows the histogram of the control effect of the TMDs with uncertainties. The small and large uncertainties are the same as those defined in Fig. 11. The sample size of the MCS is taken as 2000. The vortex-induced amplitudes at 50 s and 100 s are defined as the short-term and long-term control effects, respectively.

As shown in Fig. 15(a), the H-type TMD has the best control effect in the range of 0 to 10 mm amplitude, followed by the S-type TMD and F-

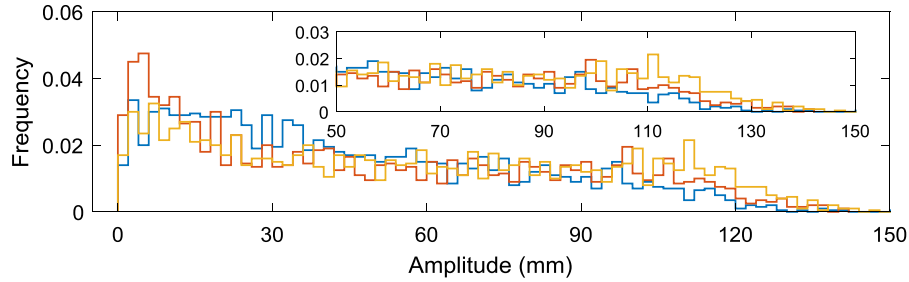
type TMD. This is because the control result with a 0–10 mm amplitude at 50 s requires a large equivalent damping ratio, indicating the TMD should focus on the effectiveness rather than the robustness. The F-type TMD has the best control effect in the range of 0 to 50 mm amplitude, and two other TMDs have the similar control effect. The MSC in the range of 50 mm to 110 mm amplitude is considered a failure in mitigating the VIV of the bridge. As shown in Fig. 15(b), the F-type and H-type TMDs have better control effect than the S-type TMD in the



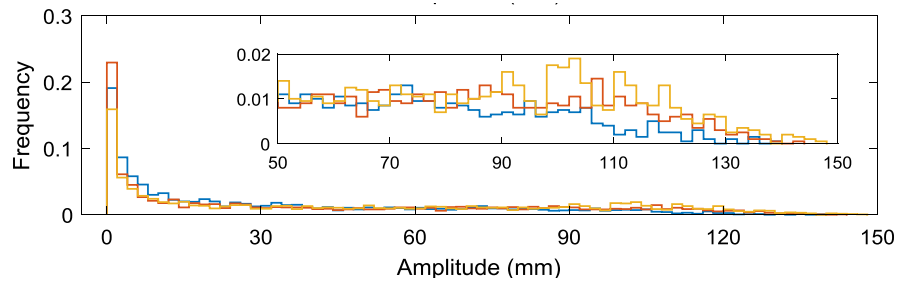
(a) Control effect at 50 s with small uncertainty



(b) Control effect at 100 s with small uncertainty



(c) Control effect at 50 s with large uncertainty



(d) Control effect at 100 s with large uncertainty

Fig. 15. Histogram of the control effect of the TMDs with uncertainties.

range of 0 to 10 mm amplitude. This is because the control result with a 0–10 mm amplitude at 100 s does not require a large equivalent damping ratio and gradually focuses on the robustness. The comparison result in the range of 50 mm to 110 mm amplitude shows that the F-type TMD and S-type TMD are the most reliable and unstable in the VIV mitigation of the bridge, respectively. In the large uncertainty case, the comparison result is the same as the that in the small uncertainty case,

as shown in Fig. 15(c) and (d). The results prove that the F-type formulas selection is the most robust and can avoid a high probability of failure in the VIV mitigation of the bridge.

Based on the robust analysis (or probability of failure analysis) in Section 4 and the MCS in Section 5, the F-type TMD in general is the best choice for the VIV mitigation of bridges considering the effectiveness and robustness simultaneously.

6. Conclusion

In this paper, we investigated the parameter determination of the TMD for mitigating the VIV in bridges. An equivalent damping ratio of the bridge is proposed to characterize the TMD contribution to the VIV mitigation based on a semi-empirical nonlinear model. The equivalent damping ratio depends only on the TMD parameters, not involving aerodynamic parameters. The result of the uncertainty analysis shows that the equivalent damping ratio is sensitive to the uncertainty in the stiffness of the bridge, while the desired damping ratio is moderate to the uncertainties in aerodynamic parameters. Based on the two damping ratios, the probability of failure of the VIV mitigation of bridge using the TMD is defined. Three typical design formulas are compared from the perspective of the effectiveness, robustness, and reliability. The S-type TMD has the most optimal control effect but with a bad robustness. The F-type TMD has the most robust control effect and avoid a high probability of failure. The design formulas for the free-vibration (the F-type design formulas) are recommended for the TMD mitigating the VIV in bridges. At last, a numerical example of the deep-water non-navigable bridge of the Hong Kong-Zhuhai-Macao Bridge confirms the rationality of the suggested design formulas.

CRediT authorship contribution statement

Jun Dai: Conceptualization, Methodology, Software, Data curation, Writing - original draft, Validation, Writing - review & editing. **Zhao-Dong Xu:** Conceptualization, Methodology, Software, Data curation, Writing - original draft, Supervision, Validation, Writing - review & editing. **Pan-Pan Gai:** Conceptualization, Methodology, Software, Data curation, Writing - original draft, Validation, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.engstruct.2020.111084>.

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